

Algebra II - Mid-Term - 80 Marks

March 13, 2025: Time: Two hours

1. Show that every group is isomorphic to a subgroup of S_n . (5 marks)
2. Let G be a group and $a, b \in G$. Show that $o(a) = o(bab^{-1})$ (5 marks)
3. Let $H = D_n$ and $G = S_n$.
 - (a) Show that H a subgroup of G . (5 marks)
 - (b) Is H a normal subgroup of G ? (5 marks)
 - (c) Let $n = 4$. Describe the distinct cosets of G/H . (5 marks)
4. Let $G = S_4$ and $Y = \{1, 2, 3, 4\}$. Let X be the set of all two element subsets of Y . Determine the stabiliser $\text{Stab}_G(X)$. (5 marks)
5. Prove or disprove: There exists a non-trivial homomorphism from the quaternion group Q_8 onto $\mathbb{Z}/2\mathbb{Z}$. (10 marks)
6. Let G be the group generated by $\left\{ \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \text{ where } a, b, c \in \mathbb{Z}_p \right\}$.
 - (a) Describe $Z(G)$. (5 marks)
 - (b) Does it have a subgroup of order p . (5 marks)

7. Let G be a non-abelian group of order 55.
- (a) Show that G is a semi-direct product of a Sylow 5-subgroup and a Sylow-11 subgroup but not a direct product. (8 marks)
 - (b) Describe all subgroups of this group. (7 marks)
8. Let G be a group. The automorphisms obtained by conjugation are called inner automorphisms denoted by $\text{Inn}(G)$
- (a) Show that $\text{Inn}(G)$ is a normal subgroup. (5 marks)
 - (b) Let $G = Q_8$, the quaternion group. Describe the elements of $\text{Inn}(G)$. Is $\text{Inn}(G)$ a cyclic group? (10 marks)